



AGGREGATED AGGREGATES

E-COMMERCE PLATFORM

Technical & Product Specification v1.0

A Specialised Operating Division of Besbpo Group

September 2026

EXECUTIVE SUMMARY

Why This Platform, and Why This Shape

This specification plans the Aggregated Aggregates e-commerce platform to the same standard already set by Roofsteel, Bricksplaza, and Aluminium Store — the three Besbpo Group materials divisions that have gone through this exact planning-to-build sequence before. It is the planning layer: it fixes scope, architecture, data model, and module design before any Claude Code scaffold or agentic build-out begins.

Aggregated Aggregates is not a high-SKU configurator business like Aluminium Store, nor a linear made-to-length business like Roofsteel. It sells graded bulk material — sub-base gravels, crushed stone, sand, ballast, decorative aggregate — priced simultaneously by weight (R/ton) and volume (R/m³), sold both loose in bulk and bagged, and delivered on a distance-banded tipper-truck logistics model. The platform's shape follows from that: a bulk/bag unit-of-sale toggle with a live tonnage-volume calculator, a distance-banded delivery engine, and an RFQ path for the large civil orders and >100km deliveries that the existing pricing framework already quotes individually rather than at a fixed rate.

The platform inherits its foundation directly from work already completed for Aggregated Aggregates: the 48-SKU product portfolio across 9 categories, the 6-sheet pricing engine (Category Markup Bands, Customer Tiers, Delivery & Additional Charges, Competitive Benchmark, Sample Priced Catalogue), the brand system (“Every Layer Starts Here”, Basalt/Limestone/Seam Blue/Ochre Gold), and the 15-slide capacity deck. Nothing in this document invents new pricing or product scope — it wires the platform to what already exists and is proven correct.

What This Document Covers

- A feature adoption matrix, scored against the pattern already refined across Roofsteel, Bricksplaza, and Aluminium Store
- Recommended architecture and technology stack, consistent with the Group's standing infrastructure preferences
- A core data model outline
- Five key-screen wireframes: homepage, category/product listing, product detail with the bulk/bag calculator, the RFQ/bulk-quote flow, and the trade account dashboard
- Six Aggregated-Aggregates-specific modules, each described in build terms
- Payment strategy, fulfilment and logistics integration, a phased build roadmap, and the confirmed decisions locking scope before Phase 1 begins

SCOPE

Feature Adoption Matrix

Roofsteel, Bricksplaza, and Aluminium Store each began with a feature-adoption pass — translating a general e-commerce blueprint into what that division's actual business model needs. Aggregated Aggregates has no separate blueprint document of its own; instead, this matrix scores its features directly against the pattern those three platforms have already converged on, adapting where the bulk/graded-material model differs from theirs.

Feature	Decision	Rationale
Multi-vendor marketplace	Drop	Single-vendor model on a ~50-strong approved partner-supplier network across South Africa — not a Takealot-style marketplace, and not Besbpo-owned yards; stock is sourced from partners rather than held as owned inventory.
Product catalogue & category browse	Adopt	Standard across all three precedent platforms.
Bulk / bag unit-of-sale toggle + tonnage-volume calculator	Adapt	Aggregated-Aggregates-specific: converts between R/ton and R/m ³ using each category's bulk density, and to R/bag using the bag premium multiplier — both already defined in the pricing engine.
Trade account & tiered pricing (Retail / Trade / Volume)	Adopt	Same 3-tier shape as Bricksplaza and Aluminium Store; rates and thresholds are Aggregated Aggregates' own (8% / 15% off list).
RFQ / bulk quote request flow	Adopt	Serves the same purpose as Roofsteel's cut/bend service selector: a request path for orders the pricing engine already flags as quote-only (≥10m ³ Volume tier, >100km delivery).
Distance-banded delivery calculator	Adapt	Own banding: 0–30km included, 30–60km and 60–100km flat add-ons by load size, >100km quoted individually — pulled live from the Delivery & Additional Charges sheet.
Compliance document attachment (SANS / COA)	Adopt	Same pattern as Roofsteel's compliance-doc module and Bricksplaza's NHBRC pass-through documents.
CMI (Contract Manufacturing Issuer) partner network	Drop	CMI answers scale-skepticism for manufactured/fabricated goods (aluminium extrusions, roll-formed steel). Aggregated Aggregates sources quarried/mined material from its ~50-strong approved partner-supplier network instead — a different trust problem with a different answer.
Made-to-Length / custom configurator	Drop	Not applicable — aggregate is not cut to length.
Business Desk / B2B portal	Adapt	A lighter trade-account dashboard (tier status, orders, standing agreements, invoices) rather than Aluminium Store's full Business Desk module suite.
CMS / admin content management	Adopt	Standard across all three precedent platforms.
Site-wide AI chat widget	Defer	Shipped on Aluminium Store; useful longer-term, not required for MVP.

Feature	Decision	Rationale
Loyalty / newsletter / promotions modules	Defer	Scope that grew on Aluminium Store; not core to an MVP for a bulk-materials buyer.
6-option SSO (Email/X/Facebook/Google/Apple/Microsoft)	Adapt	Reduce to Email + Google + Microsoft — this buyer base (contractors, civil buyers) skews B2B; consumer-social logins are low-value here.
Supplier locator	Adopt	Maps directly to the ~50-supplier partner network model rather than an owned-yard footprint.
Freight / courier integration (Besfleet + external)	Adopt	Confirmed hybrid model: Besfleet alongside 15+ external tipper-truck delivery partners from launch — a broader external network than Roofsteel is running at this stage.

ARCHITECTURE

Recommended Architecture & Technology Stack

This stack is consistent with the Group's standing infrastructure preferences and with the pattern already built and running for Roofsteel and Bricksplaza — it is not a fresh choice per division.

Layer	Choice	Notes
Frontend	Next.js (App Router, TypeScript, Tailwind)	Same framework as Roofsteel, Bricksplaza, Aluminium Store.
Backend	NestJS, modular monolith	Auth, catalogue, orders, trade-accounts, quotes/RFQ, CMS as separate modules.
Pricing microservice	FastAPI	Cross-checked against the Aggregated Aggregates pricing framework xlsx to the cent, the same QA standard used on Bricksplaza and Aluminium Store.
Database	PostgreSQL via Supabase	Group standard.
Caching / queues	Upstash Redis + BullMQ	Group standard.
Hosting	Render	Confirmed — matches the Roofsteel and Bricksplaza platform pattern.
Infra / IaC	AWS af-south-1, Terraform/Terragrunt, ArgoCD GitOps	Group standard, data sovereignty requirement.
Domain split	aggregates.store + app.aggregates.store	Confirmed domain — same corporate-site-plus-app-subdomain split as Roofsteel and Bricksplaza.

DATA MODEL

Core Data Model

Model names follow the Prisma convention used on the Roofsteel and Bricksplaza platforms. This is the core set; supporting models (sessions, audit logs, media assets) are omitted here for brevity but follow the same pattern established on those builds.

Model	Purpose
User / OAuthAccount	Authentication and account linking (Email, Google, Microsoft).
Company	Trade account entity — holds the customer tier, standing agreements, and delivery addresses.
Product / Category / ProductImage	The 48-SKU / 9-category portfolio and its imagery.
PriceBand	Category-level bulk density, base cost, markup %, and bag premium multiplier — mirrors the Category Markup Bands sheet.
CustomerTier	Retail / Contractor-Trade / Volume-Civil-Bulk, with each tier's discount off list.
DeliveryBand	Distance × load-size matrix (0–30 / 30–60 / 60–100 / >100km × 6m ³ / 10m ³ / 14m ³ +).
SupplierLocation / DeliveryPoint	The ~50-strong approved partner-supplier network a delivery band is measured from; there is no owned-yard entity in this model.
Quote / QuoteLineItem	The RFQ / bulk-quote flow — captures quote-only orders before they become a priced Order.
Order / OrderLineItem	Confirmed orders, unit-of-sale (bulk/bag) and quantity recorded per line.
Shipment	Delivery tracking — carrier (Besfleet or external), distance band, and delivery fee applied.
ComplianceDocument	SANS / COA (Certificate of Analysis) attachments per product or batch.
Invoice / Statement	Billing records, including PO-based billing for Volume-tier civil accounts.

INTERFACE

Key Screen Wireframes

Five screens carry the platform's core logic — the storefront, the calculator that makes bulk/bag pricing legible, the RFQ path for quote-only orders, and the trade account view. These are low-fidelity layout wireframes: they fix structure and content hierarchy, not visual design — the brand system (Basalt/Limestone/Seam Blue/Ochre Gold, Space Grotesk/Inter/IBM Plex Mono) is applied at high-fidelity design/build stage.

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WIREFRAME 01 — Homepage
Corporate + storefront hybrid landing page

Logo
Products
Trade Accounts
Delivery Areas
About
Contact
[Get a Quote]
[Sign In]

SANS / COLTO GRADED

[hero: rotating banner — featured product imagery]

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Sub-base, crushed stone, sand & decorative aggregate

Shop Products

Request a Quote

QUICK TONNAGE CALCULATOR

Product ▼

Quantity (m³ / tons)

Delivery Postcode

[Get Estimate]

SHOP BY CATEGORY

Sub-Base & Base Course

Crushed Stone

Sand & Fine Aggregates

Crusher Run

Ballast & Rail

Drainage Stone

Decorative & Landscaping

Agricultural & Recycled

● SANS 1200 / 1083 Graded
● KZN + Gauteng Yard Network
● Trade Accounts Available
● Besbpo Group Division

Footer — group ecosystem links, contact, compliance, social

1. Homepage — corporate + storefront hybrid, with a quick tonnage calculator surfaced above the fold.

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WIREFRAME 02 — Category & Product Listing

Filterable grid with unit-of-sale toggle

Home / Products / Crushed Stone Aggregates

FILTERS

Category ▾

Grading / Size ▾

Unit of Sale (Bulk / Bag) ▾

Segment (Civil/Comm/Ind/Res) ▾

Delivery Radius ▾

Apply Filters

48 products

Sort: Relevance ▾

IMG

G5 Natural Gravel

R400.00 /ton • R580.00 /m³

Bulk / Bag ▾

Add to Quote

IMG

19mm Crushed Stone (Dolomite)

R400.00 /ton • R580.00 /m³

Bulk / Bag ▾

Add to Quote

IMG

River Sand (Washed)

R400.00 /ton • R580.00 /m³

Bulk / Bag ▾

Add to Quote

IMG

Crusher Run 0-19mm

R400.00 /ton • R580.00 /m³

Bulk / Bag ▾

Add to Quote

IMG

Rail Ballast

R400.00 /ton • R580.00 /m³

Bulk / Bag ▾

Add to Quote

IMG

French Drain Stone

R400.00 /ton • R580.00 /m³

Bulk / Bag ▾

Add to Quote

IMG

River Pebble

R400.00 /ton • R580.00 /m³

Bulk / Bag ▾

Add to Quote

IMG

Agricultural Lime

R400.00 /ton • R580.00 /m³

Bulk / Bag ▾

Add to Quote

IMG

Recycled Crushed Concrete (RCA)

R400.00 /ton • R580.00 /m³

Bulk / Bag ▾

Add to Quote

2. Category & product listing — filterable grid with a bulk/bag toggle on every card.

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WIREFRAME 03 — Product Detail — Bulk/Bag Calculator

Unit toggle drives live price + delivery-band estimate

Home / Products / Crushed Stone Aggregates / 19mm Crushed Stone (Dolomite)

**19mm Crushed Stone (Dolomite)**

Crushed Stone Aggregates • SANS 1083

R400.00 / ton

R580.00 / m³ • R11.78 / 20kg bag

TONNAGE / BAG CALCULATOR**Unit of Sale** (☒ Bulk (ton/m³) ☐ Bagged)**Quantity** [10] m³ ⇄ [14.5] tons**Customer Tier** Retail ▼ (sign in for Trade/Volume pricing)**Delivery Distance** [Postcode/Address] → 32km band**Subtotal: R5,800.00 + Delivery: R450.00****Add to Quote****Request Bulk RFQ****Specification**

Grading Curve

Compliance Docs (SANS/COA)

Delivery & Returns

Tab content — grading curve chart / SANS 1083 spec table

3. Product detail — the tonnage/bag calculator, live-priced against tier and delivery distance.

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WIREFRAME 04 — RFQ / Bulk Quote Request
Multi-step flow for civil bulk & >100km delivery quotes

1. Project Details
2. Materials & Quantities
3. Delivery Location
4. Review & Submit

STEP 2 — Materials & Quantities

Product	Grading/Size	Unit	Quantity	Est. Unit Price	
G3 Crushed Stone	G3 (COLTO)	m³	120	R580.00	×
13.2mm Crushed Stone	13.2mm	ton	45	R400.00	×
Ferrocrite Ballast	Graded blend	ton	80	R256.20	×
+ Add Another Product					

Note: orders ≥ 10m³ (Volume/Civil Bulk tier) or delivery > 100km are quoted individually — our team responds within 1 business day with delivered pricing.

← Back
Continue →

4. RFQ / bulk quote request — the path for Volume-tier and >100km orders that the pricing engine quotes individually.

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WIREFRAME 05 — Trade Account Dashboard

Customer portal — tier status, orders, standing agreements

Dashboard

Order History

Quotes / RFQs

Standing Agreements

Delivery Addresses

Invoices & Statements

Account Settings

Welcome back — Company: XYZ Civil Contractors (Pty) Ltd

ACCOUNT TIER

Contractor / Trade
(8% off list)

ORDERS THIS MONTH

6 orders

OUTSTANDING BALANCE

R48,320.00

STANDING AGREEMENT

KZN Civil — active

RECENT ORDERS

Order #	Date	Products	Delivery Yard	Status	Total
AA-10245	18 Sep 2026	G5 Gravel, 19mm Stone	Pinetown Yard	Delivered	R14,200
AA-10238	11 Sep 2026	River Sand x 10m³	Pinetown Yard	In Transit	R5,555
AA-10221	29 Aug 2026	Crusher Run 0-19mm	Germiston Yard	Delivered	R8,960

5. Trade account dashboard — tier status, recent orders, and standing agreements.

MODULES

Aggregated-Aggregates-Specific Modules

1. Bulk / Bag Toggle + Tonnage-Volume Calculator

The single feature that makes the pricing engine usable on a storefront. Every product carries a bulk density (kg/m³) and, where sold bagged, a bag premium multiplier — both already defined in the Category Markup Bands sheet. The calculator converts live between tons and m³ for bulk orders, and computes the bagged price from the same base rate, so the displayed price is never a second, hand-maintained number.

2. Distance-Banded Delivery Calculator

Reads the Delivery & Additional Charges sheet directly: 0–30km from the nearest approved partner-supplier location is included, 30–60km and 60–100km carry flat add-ons banded by load size (6m³ / 10m³ / 14m³+ Interlink), and anything beyond 100km routes to the RFQ flow rather than showing a number the business hasn't priced.

3. RFQ / Civil Bulk Quote Flow

A multi-step request — project details, materials and quantities, delivery location, review — for orders the pricing framework already treats as quote-only: Volume/Civil Bulk tier orders ($\geq 10\text{m}^3$) and any delivery beyond 100km. This is the same design answer Roofsteel gave to its cut/bend service selector: some orders are too variable to price at checkout, so the platform collects the right detail and routes it to a human quote instead of guessing.

4. Trade Account & Customer Tier Management

Retail (list price), Contractor/Trade (8% off), and Volume/Civil Bulk (15% off) — the same three tiers already defined in the pricing engine's Customer Tiers sheet. A registered company account carries its tier, standing delivery addresses, and (for Volume accounts) any standing supply agreement, so repeat civil buyers aren't re-entering the same delivery and billing detail on every order.

5. Compliance Document Attachment (SANS / COA)

Every product category maps to a reference standard — SANS 1200-G for G1–G10 grading, SANS 1083 for concrete aggregate, COLTO/TRH14 for road-building material. This module attaches the relevant SANS reference and, where available, a batch-specific Certificate of Analysis to the product page and the order record — the same pattern used for Roofsteel's compliance documents and Bricksplaza's NHBRC pass-through.

6. Supplier & Delivery-Point Locator

Surfaces the ~50-strong approved partner-supplier network the delivery calculator measures distance from — a broker model with no owned yards. Launches on the KZN + Gauteng footprint and extends to the Group's standard 7-province footprint as the network grows.

PAYMENTS

Payment Strategy

Method	Use Case	Status
PayFast	Retail card / EFT checkout	Adopt — default across all Group e-commerce platforms.
Lulapay	B2B buy-now-pay-later for Contractor/Trade accounts	Adopt — same partner already onboarding for Roofsteel.
PayJustNow	Retail-tier instalment option	Adopt, retail tier only — consistent with Roofsteel.
EFT / Purchase Order	Volume/Civil Bulk tier, standing agreements	Adopt — large civil buyers typically settle by PO/invoice on net terms, not card; this is Aggregated Aggregates' own addition to the payment mix given its Volume-tier profile.
Yoco	—	Defer, matching the Roofsteel decision.

LOGISTICS**Fulfilment & Logistics Integration**

- Freight is calculated from real weight/volume data and the distance-banded matrix already defined in the pricing engine — not a flat per-item shipping rate, which doesn't hold for tipper-truck bulk delivery.
- Besfleet, the Group's own trucking division, runs alongside a network of 15+ external tipper-truck delivery partners from launch — a hybrid courier model confirmed for this division.
- Fulfilment launches on the KZN + Gauteng partner-supplier network, expanding to the Group's standard 7-province footprint as the ~50-supplier network grows nationally.
- Load-size logic (6m³ / 10m³ / 14m³+ Interlink) is built into both the checkout delivery estimate and the RFQ flow, since a bulk tipper order is priced by truckload, not by parcel.

DELIVERY PLAN

Phased Build Roadmap

Follows the same five-phase HITLAD (human-in-the-loop agentic development) pattern used to plan Roofsteel's build: Claude Code scaffolds the foundation, an agentic build-out fills in each module, and every phase closes on human PR review before the next begins.

Phase	Scope
1. Foundation	Schema, auth (Email/Google/Microsoft), category and product structure, base storefront scaffold.
2. Pricing & Calculators	Pricing microservice wired to the Category Markup Bands and Customer Tiers sheets; bulk/bag calculator; distance-banded delivery calculator.
3. Trade & Quote Flows	Trade account/tier management, RFQ/bulk-quote flow, compliance document attachment.
4. Storefront & CMS	Full storefront polish, CMS/admin, supplier & delivery-point locator.
5. QA & Launch Prep	Human PR review throughout; pricing microservice reconciled against the pricing xlsx to the cent, matching the QA bar set on Bricksplaza and Aluminium Store.

LoC estimate: given Aggregated Aggregates' bulk/commodity-materials scope — closer to Roofsteel's model than to Aluminium Store's high-SKU configurator scope — a working benchmark of roughly 30,000–38,000 lines of code is reasonable, anchored to Roofsteel's real delivered figure of 35,521 LoC.

BEFORE PHASE 1

Confirmed Decisions

Signed off by Fortune ahead of Phase 1 — these replace what was open scope in the working draft of this spec.

- Supplier network: stock is sourced from Aggregated Aggregates' own partner network — roughly 50 approved suppliers across South Africa — rather than an existing personal-capacity relationship (Roofsteel's pattern) or a network built from scratch (Bricksplaza's pattern).
- Stock model: broker/network model confirmed. Aggregated Aggregates holds no owned inventory at owned yards, so StockLevel/StockMovement tracking is dropped from the data model — SupplierLocation/DeliveryPoint carries the network instead.
- Domain: aggregates.store, with the same corporate-site-plus-app-subdomain split used on Roofsteel and Bricksplaza (app.aggregates.store).
- VAT invoicing: bills under the Besbpo Group company VAT registration from day one, consistent with Aggregated Aggregates operating as a Group division rather than a separately VAT-registered entity.
- Hosting: Render, matching the Roofsteel/Bricksplaza precedent.
- Courier: hybrid model confirmed — Besfleet alongside 15+ external tipper-truck delivery partners from launch.
- Launch geography: KZN + Gauteng first, matching Bricksplaza's sequence, expanding to the Group's standard 7-province footprint as the partner-supplier network grows.